Excel Project on Business KPIs and Reporting Dashboard

Five-Week Task Report: Data Cleaning, Analytical Planning, Visualization, Statistical Analysis and Dashboard Development

Dataset: Excel Sales Dashboard (SalesDataTable, 2026)

Tool: Microsoft Excel (Pivot Tables, Pivot Charts, Slicers, Map Chart)

Contents

Week	Task	Main output
Week 1	Data Cleaning and Preparation with Advanced Excel Techniques	Cleaned SalesDataTable with Year, Month No, Month and Quarter helper columns
Week 2	Analytical Planning and Strategic Data Analysis	KPI framework, analysis workflow, timeline and KPI pivot tables
Week 3	Data Visualization and Reporting with Advanced Excel	Line, bar, pie, donut and map charts with rationale and insights
Week 4	Advanced Statistical Analysis and Scenario Modeling	Descriptive and inferential statistics, what-if scenarios, sensitivity table
Week 5	Dashboard Development and Final Reporting	Interactive KPI dashboard with slicers and report connections

How to read this report. Every week is documented against its task: Objective, Task Overview, Key Steps, Deliverables and Evaluation focus. Screenshots are taken from the working Excel file. Where a task required material that the original notes did not cover (for example validation rules or scenario analysis), it has been added and is marked **(Added)**. All figures come from the dashboard totals visible in the screenshots.

Project snapshot

KPI	Value	How it is calculated
Total Revenue	$645,223	Sum of Revenue
Total Gross Profit	$296,384	Sum of Gross Profit
Gross Profit Margin	45.94%	Gross Profit / Revenue
Total Units Sold	6,698	Sum of Units Sold
Total Orders	2,707	Count of ID
COGS	$348,839	Sum of COGS

WEEK 1

Data Cleaning and Preparation with Advanced Excel Techniques

Objective: transform raw sales data into a clean, analyzable dataset and document the logic behind every step.

1. Task Overview

This task simulates a real-world situation in which raw data is incomplete, inconsistent or not yet structured for analysis. The raw sales records were stored in an Excel Table named **SalesDataTable** with the columns ID, Order Date, Region, Country, Sales Channel, Customer Type, Product Category, Product, Units Sold, Unit Price, Discount, Unit Cost and Revenue. The raw table had no Year, Month or Quarter fields, so time-based analysis was not possible. The cleaning and preparation work below fixes that and documents each decision.

2. Potential Data Issues and Cleaning Plan

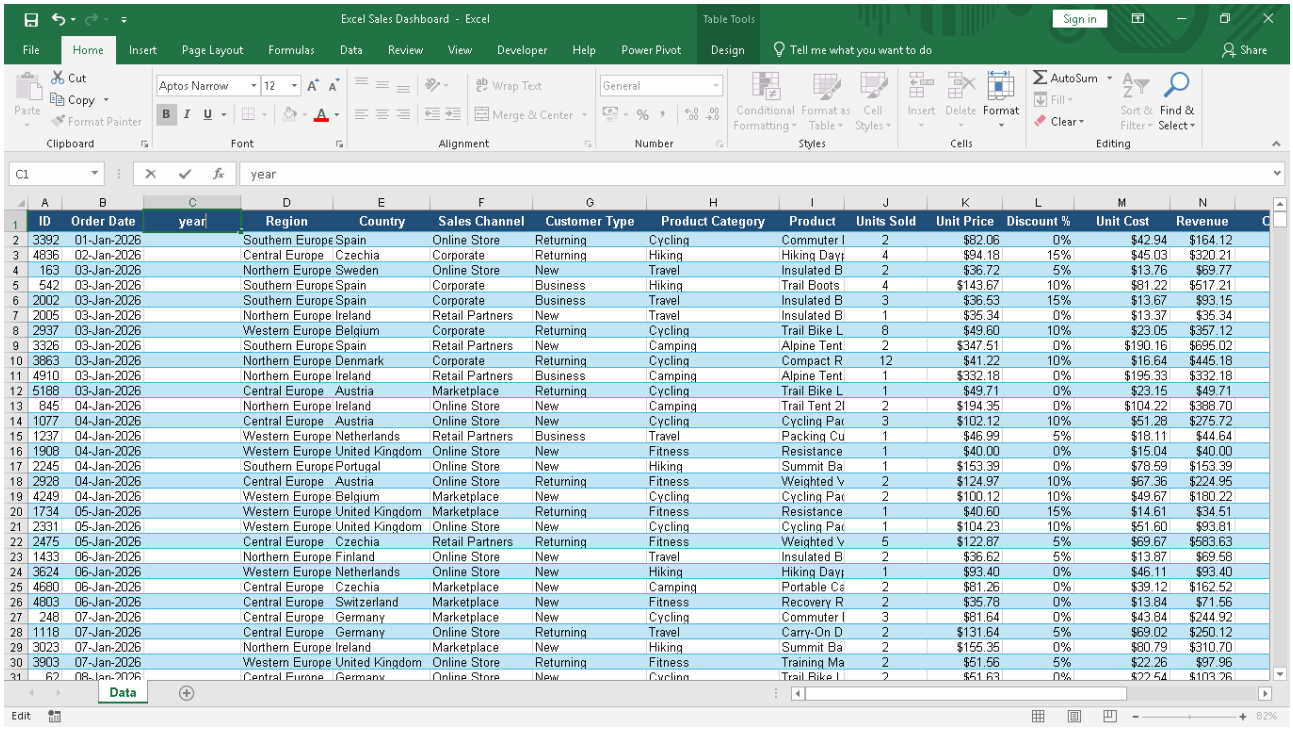
Potential issue	Why it matters	Technique used
No Year / Month / Quarter field	Pivot tables cannot group by period without helper columns	YEAR, MONTH, TEXT and ROUNDUP formulas in Table columns
Wrong number format after formula	YEAR() result displayed as a date (e.g. 18-Jul-1905)	Format Cells -> Number -> General
Months sorting alphabetically	Apr, Aug, Dec ... instead of Jan, Feb, Mar	Pivot Table More Sort Options -> Manual / Custom list order
Inconsistent text, spaces, duplicates **(Added)**	Creates duplicate categories in pivots and slicers	TRIM, PROPER, Remove Duplicates
Missing or invalid values **(Added)**	Wrong totals and margins	IF, ISBLANK, IFERROR, Data Validation, Conditional Formatting

3. Step-by-Step Cleaning Process

Because the data sits in an Excel Table, each formula below is entered once and auto-fills the whole column using structured references such as `[@[Order Date]]`.

Step 1: Inspect the raw data

The sales table was opened and checked: 13 columns, one row per transaction, 2,707 records. A new column named **year** was inserted next to Order Date.



ID	Order Date	year	Region	Country	Sales Channel	Customer Type	Product Category	Product	Units Sold	Unit Price	Discount %	Unit Cost	Revenue
3392	01-Jan-2026		Southern Europe	Spain	Online Store	Returning	Cycling	Commuter I	2	$82.06	0%	$42.94	$164.12
4836	02-Jan-2026		Central Europe	Czechia	Corporate	Returning	Hiking	Hiking Day	4	$94.18	15%	$45.03	$320.21
163	03-Jan-2026		Northern Europe	Sweden	Online Store	New	Travel	Insulated B	2	$36.72	5%	$13.76	$69.77
542	03-Jan-2026		Southern Europe	Spain	Corporate	Business	Hiking	Trail Boots	4	$143.67	10%	$81.22	$517.21
2002	03-Jan-2026		Southern Europe	Spain	Corporate	Business	Travel	Insulated B	3	$36.53	15%	$13.67	$93.15
2005	03-Jan-2026		Northern Europe	Ireland	Retail Partners	New	Travel	Insulated B	1	$35.34	0%	$13.37	$35.34
2937	03-Jan-2026		Western Europe	Belgium	Corporate	Returning	Cycling	Trail Bike L	8	$49.60	10%	$23.05	$357.12
3326	03-Jan-2026		Southern Europe	Spain	Retail Partners	New	Camping	Alpine Tent	2	$347.51	0%	$190.16	$695.02
3863	03-Jan-2026		Northern Europe	Denmark	Corporate	Returning	Cycling	Compact R	12	$41.22	10%	$16.64	$445.18
4910	03-Jan-2026		Northern Europe	Ireland	Retail Partners	Business	Camping	Alpine Tent	1	$332.18	0%	$195.33	$332.18
5188	03-Jan-2026		Central Europe	Austria	Marketplace	Returning	Cycling	Trail Bike L	1	$49.71	0%	$23.15	$49.71
845	04-Jan-2026		Northern Europe	Ireland	Online Store	New	Camping	Trail Tent 2I	2	$194.35	0%	$104.22	$388.70
1077	04-Jan-2026		Central Europe	Austria	Online Store	New	Cycling	Cycling Pa	3	$102.12	10%	$51.28	$275.72
1237	04-Jan-2026		Western Europe	Netherlands	Retail Partners	Business	Travel	Packing Cu	1	$46.99	5%	$18.11	$44.64
1908	04-Jan-2026		Western Europe	United Kingdom	Online Store	New	Fitness	Resistance	1	$40.00	0%	$15.04	$40.00
2245	04-Jan-2026		Southern Europe	Portugal	Online Store	New	Hiking	Summit Ba	1	$153.39	0%	$78.59	$153.39
2928	04-Jan-2026		Central Europe	Austria	Online Store	Returning	Fitness	Weighted V	2	$124.97	10%	$67.36	$224.95
4249	04-Jan-2026		Western Europe	Belgium	Marketplace	New	Cycling	Cycling Pa	2	$100.12	10%	$49.67	$180.22
1734	05-Jan-2026		Western Europe	United Kingdom	Marketplace	Returning	Fitness	Resistance	1	$40.60	15%	$14.61	$34.51
2331	05-Jan-2026		Western Europe	United Kingdom	Online Store	New	Cycling	Cycling Pa	1	$104.23	10%	$51.60	$93.81
2475	05-Jan-2026		Central Europe	Czechia	Retail Partners	Returning	Fitness	Weighted V	5	$122.87	5%	$69.67	$583.63
1433	06-Jan-2026		Northern Europe	Finland	Online Store	New	Travel	Insulated B	2	$36.62	5%	$13.87	$69.58
3624	06-Jan-2026		Western Europe	Netherlands	Online Store	New	Hiking	Hiking Day	1	$93.40	0%	$46.11	$93.40
4680	06-Jan-2026		Central Europe	Czechia	Marketplace	New	Camping	Portable Ca	2	$81.26	0%	$39.12	$162.52
4803	06-Jan-2026		Central Europe	Switzerland	Marketplace	New	Fitness	Recovery R	2	$35.78	0%	$13.84	$71.56
248	07-Jan-2026		Central Europe	Germany	Marketplace	New	Cycling	Commuter I	3	$81.64	0%	$43.84	$244.92
1118	07-Jan-2026		Central Europe	Germany	Online Store	Returning	Travel	Carry-On D	2	$131.64	5%	$69.02	$250.12
3023	07-Jan-2026		Northern Europe	Ireland	Marketplace	New	Hiking	Summit Ba	2	$155.35	0%	$80.79	$310.70
3903	07-Jan-2026		Western Europe	United Kingdom	Online Store	Returning	Fitness	Training Ma	2	$51.66	5%	$22.26	$97.96
62	08-Jan-2026		Central Europe	Germany	Online Store	New	Cycling	Trail Bike I	2	$51.63	0%	$22.54	$103.26

Figure 1.1 - SalesDataTable with the new helper column being added next to Order Date

Step 2: Create the Year column

The YEAR function extracts the year from each order date.

Column	Formula	Purpose
year	`=YEAR([@[Order Date]])`	Extract the year as a number

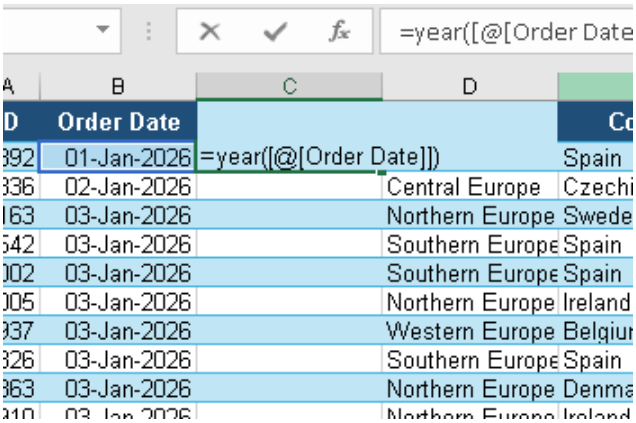


Figure 1.2 - YEAR formula entered in the first data row

Step 3: Fix the number format (before and after)

Excel automatically applied a date format to the result, so the year appeared as dates such as 18-Jul-1905 (this is how Excel displays the serial number 2026 as a date). The column was selected, **Format Cells -> Number -> General** was chosen, and the values then displayed correctly as 2026.

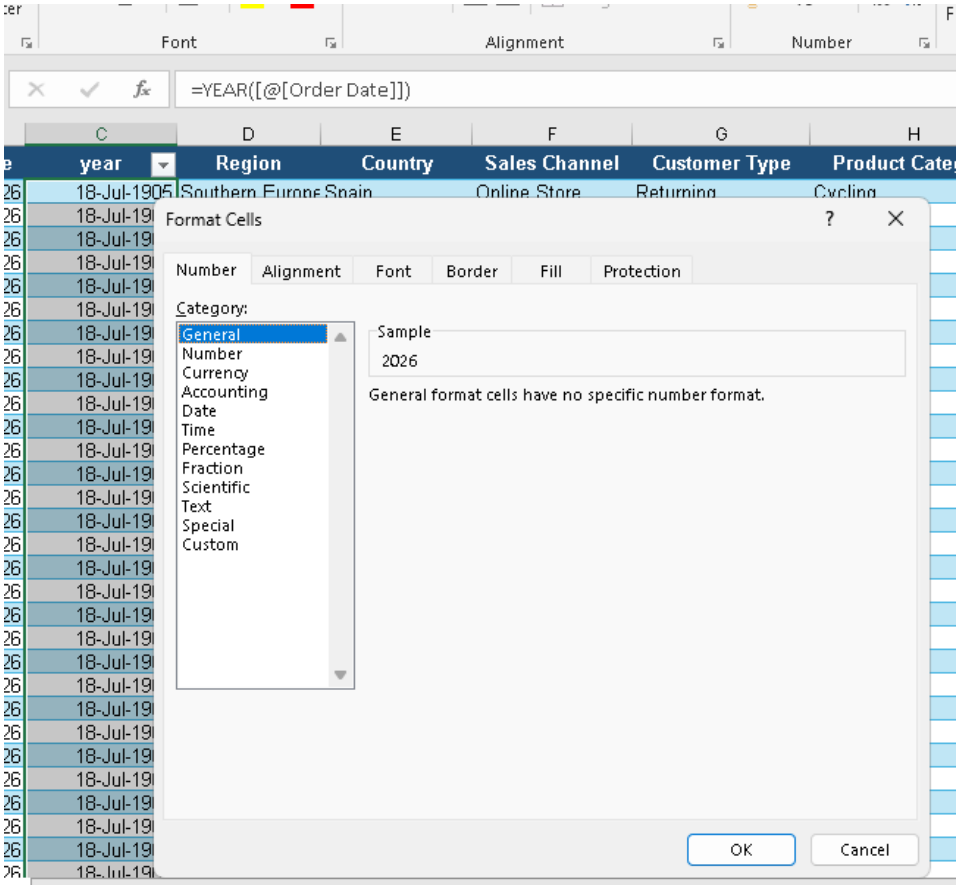


Figure 1.3 - BEFORE: year column showing dates (1905). Format Cells dialog set to General to correct it

Step 4: Create Month No, Month and Quarter columns

Column	Formula	Result / purpose
Month No	`=MONTH([@[Order Date]])`	1 to 12; used to sort months correctly
Month	`=TEXT([@[Order Date]],"MMM")`	Jan, Feb, Mar ... for chart axes
Quarter	`="Q"&ROUNDUP(MONTH([@[Order Date]])/3,0)`	Q1 to Q4 for quarterly grouping

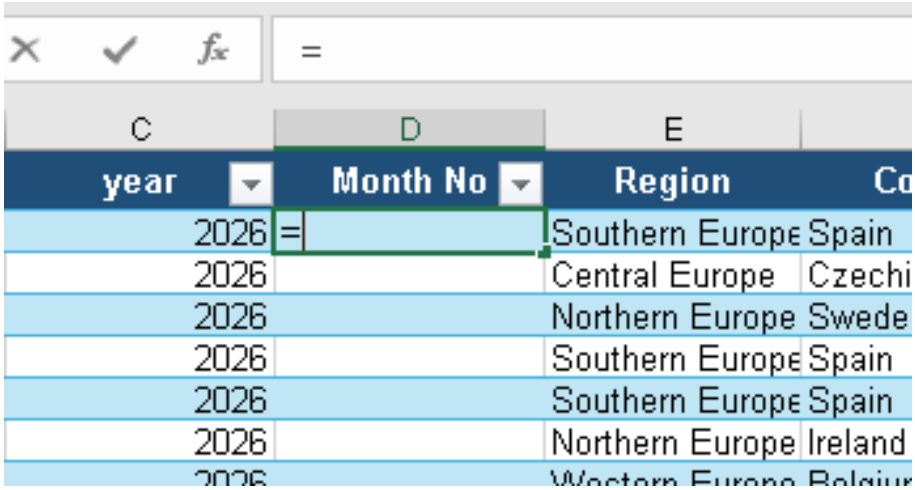


Figure 1.4 - Month No column added, ready for formula

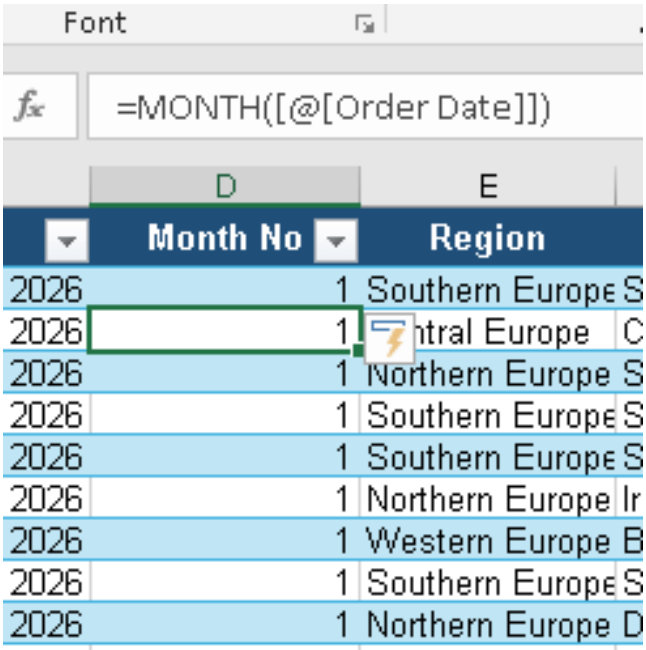


Figure 1.5 - MONTH function returns the month number

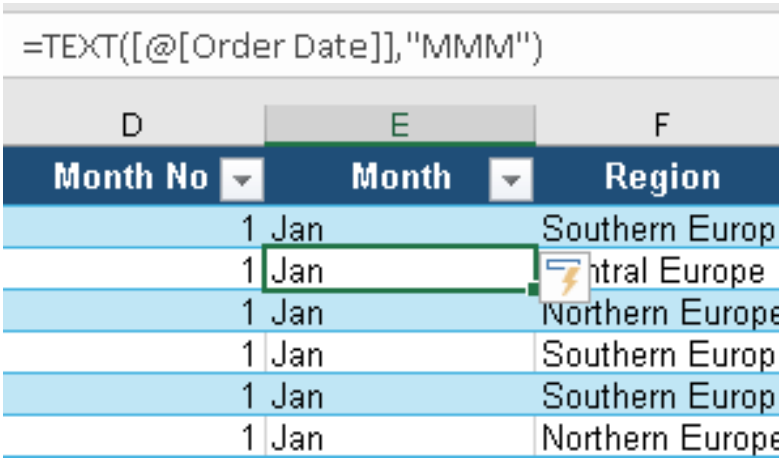


Figure 1.6 - TEXT function with the MMM format returns the month name

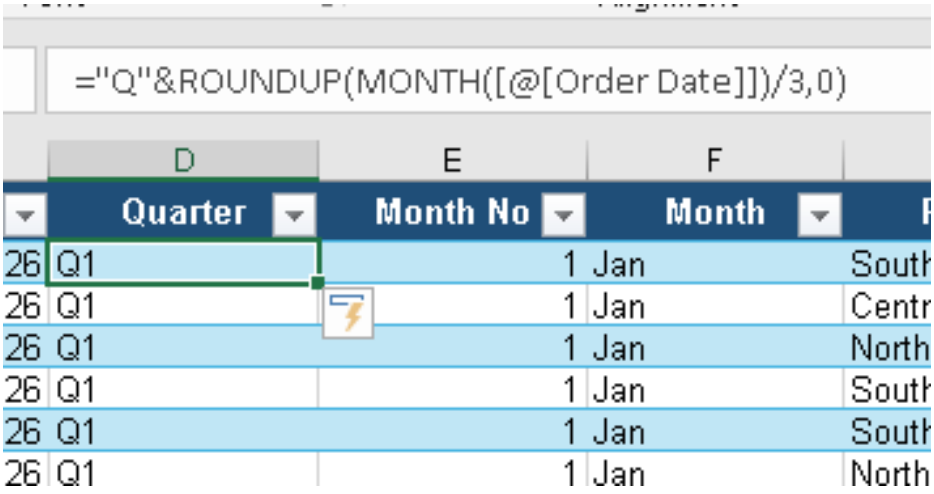


Figure 1.7 - Quarter column built with ROUNDUP and MONTH, giving Q1 to Q4

4. Data Validation and Conditional Formatting (Added)

To keep the dataset reliable as new records are appended, the following controls are recommended and were designed against the existing columns:

Control	Setting	Logic
Data Validation: Units Sold	Whole number, between 1 and 10,000	Blocks zero, negative or decimal quantities
Data Validation: Discount	Decimal between 0 and 1	Prevents discount above 100%
Data Validation: Sales Channel	List: Corporate, Marketplace, Online Store, Retail Partners	Stops misspelt channel names that would split pivot rows
Conditional Formatting: blanks	`=ISBLANK(A2)`	Highlights missing cells in red
Conditional Formatting: duplicates	Highlight Cells Rules -> Duplicate Values on ID	Flags repeated transaction IDs
Text cleanup helper	`=PROPER(TRIM([@Country]))`	Removes extra spaces and fixes capitalization
Lookup check	`=IFERROR(INDEX(Regions[Region],MATCH([@Country],Regions[Country],0)),"Check")`	Confirms every country maps to a region; shows Check if not
Revenue check	`=IF([@Revenue]=[@[Units Sold]]*[@[Unit Price]]*(1-[@Discount]),"OK","Review")`	Verifies revenue equals units x price after discount (adjust if revenue is defined differently)

5. Readiness for Further Analysis

- Dates are now split into Year, Month No, Month and Quarter, so every pivot table can group by period.
- Month No allows months to be sorted in calendar order (Jan to Dec) rather than alphabetically.
- Because the data is an Excel Table, new rows inherit the formulas automatically and pivot tables refresh without changing ranges.
- Validation rules and checks protect the quality of future data entries.
- The cleaned table is the single source for every KPI, chart and slicer in Weeks 2 to 5.

Evaluation focus: clear documentation (each formula is explained), accurate techniques (YEAR, MONTH, TEXT, ROUNDUP, IFERROR, INDEX/MATCH) and a logical flow from raw data to analysis-ready data.

WEEK 2

Analytical Planning and Strategic Data Analysis

Objective: build a data analysis plan with objectives, KPIs, workflow, timeline and expected outcomes.

1. Task Overview

This week represents the first stage of a real analysis project, where goals and methods are planned before any chart is drawn. The plan below covers the role of analysis in decision-making, the KPIs chosen and why, the dataset criteria, the step-by-step Excel process and the outcomes expected.

2. Role of Data Analysis in Decision-Making

- **Understand performance:** how much revenue and profit the business generates and how it changes by month.
- **Find where value comes from:** which countries, regions, channels and product categories contribute most.
- **Support actions:** allocate marketing budget, adjust pricing, and focus on strong markets and channels.
- **Monitor continuously:** a dashboard lets managers track KPIs without rebuilding reports.

Common analytical goals for this project: measure business size, measure profitability, identify top markets, compare channels, detect seasonality and compare regions.

3. KPIs and Strategic Rationale

KPI	Why it matters
Revenue	Measures business size
Gross Profit	Shows profit generated before operating expenses
Gross Margin %	Shows how efficiently sales generate profit
Units Sold	Separates sales volume from pricing effects
Sales Records	Shows transaction activity

Figure 2.1 - KPI framework: each KPI and why it matters

KPI	Excel definition	Strategic rationale
Revenue	Sum of Revenue	Shows overall business size and is the base for share calculations

KPI	Excel definition	Strategic rationale
Gross Profit	Sum of Gross Profit	Shows profit generated before operating expenses
Gross Margin %	`=Gross Profit / Revenue`	Shows how efficiently sales generate profit; falls when discounts or costs rise
Units Sold	Sum of Units Sold	Separates sales volume from pricing effects
Sales Records (Orders)	Count of ID	Shows transaction activity and allows average order value
COGS **(Added)**	Sum of COGS	Explains the gap between revenue and gross profit

4. Dataset Criteria and Sample Metrics

Criterion	Requirement	Met by SalesDataTable
Granularity	One row per transaction	Yes, 2,707 records
Time dimension	Date field to analyse trends	Order Date plus Year, Month, Quarter helpers
Geography	Country and region	17 countries across 4 regions
Categories	Channel, customer type, product category	4 sales channels and several product categories
Numeric measures	Units, price, cost, revenue, profit	Yes

Sample derived metrics: average revenue per unit = $96.33; average cost per unit = $52.08; average order value = $238.35; units per order = 2.47; gross margin = 45.94%.

5. Hypotheses to Test

- H1: Revenue is concentrated in a few countries. (Top 5 countries generate $296,507, about 46% of revenue.)
- H2: The Online Store is the largest sales channel. (Confirmed in Week 3: about 40% of revenue.)
- H3: Revenue follows a seasonal pattern with a mid-year peak. (Examined with the monthly line chart.)
- H4: Product category affects revenue and profit, so slicers should allow filtering by category.

6. Analysis Workflow

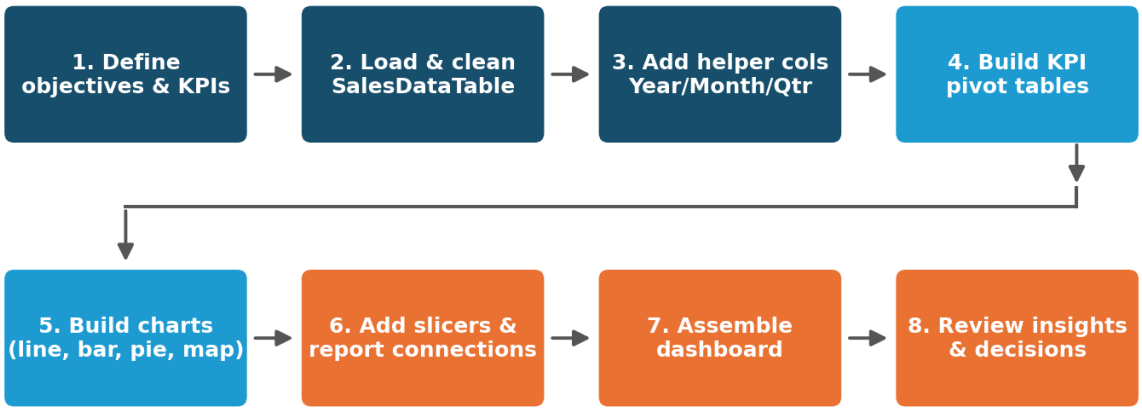


Figure 2.2 - Analysis process flowchart from objectives to decisions

7. Step-by-Step Analytical Process and Excel Functions

Step	Excel feature	Expected outcome
Create KPI pivot tables	Insert -> PivotTable -> New Worksheet (from SalesDataTable)	Revenue, Gross Profit, Units and Order count
Calculate margin	`=C4/A4`	Gross profit margin 0.459351522 (45.94%)
Pivot by Month	Rows: Month, Values: Revenue and Gross Profit	Monthly trend table
Pivot by Country	Value Filters -> Top 10 -> 5	Top 5 markets
Pivot by Channel / Region	Rows: Sales Channel or Region	Revenue share
Cross-filter	Slicers with Report Connections	One click updates all charts
Advanced options **(Added)**	`=SUMIFS()` `=XLOOKUP()` `=UNIQUE()` `=SORTBY()`	Dynamic array and conditional summaries to validate pivot results

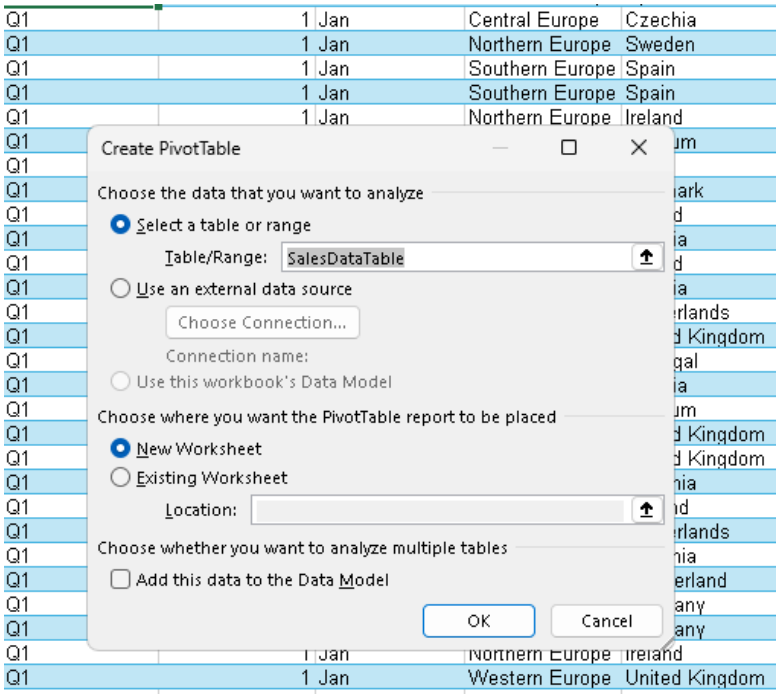


Figure 2.3 - Creating the PivotTable from SalesDataTable on a new worksheet

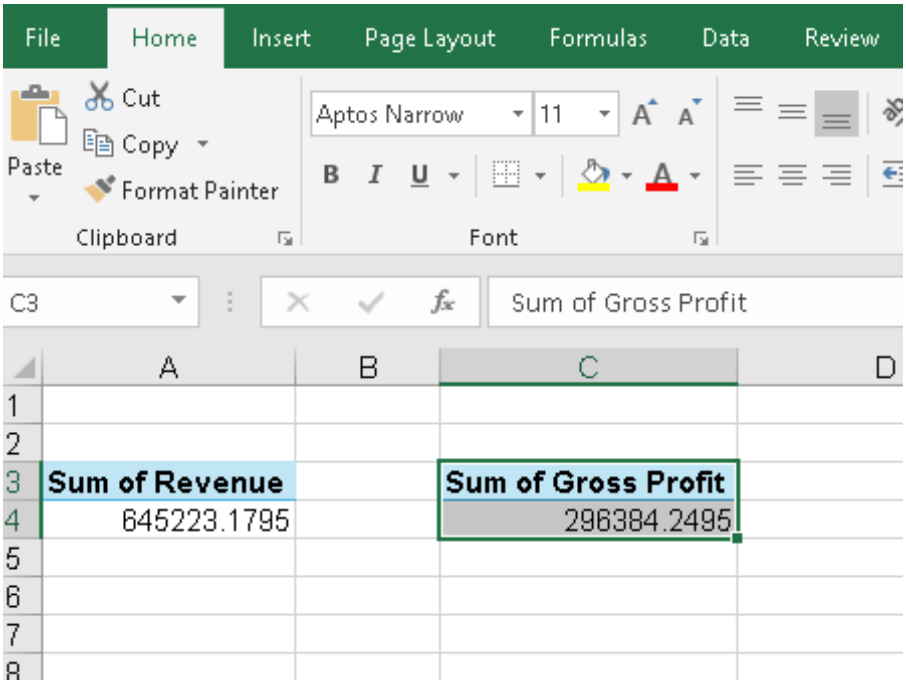


Figure 2.4 - KPI pivot: Sum of Revenue and Sum of Gross Profit

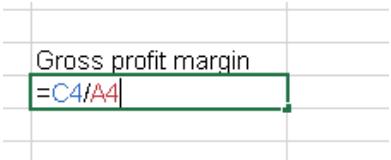


Figure 2.5 - Gross profit margin formula =C4/A4

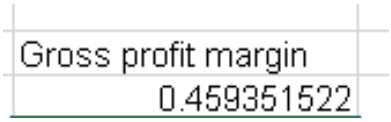
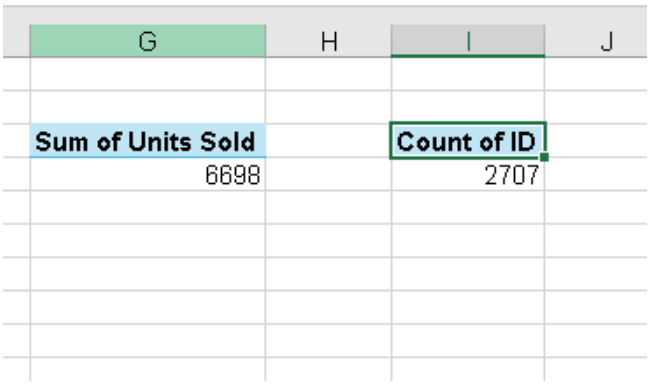
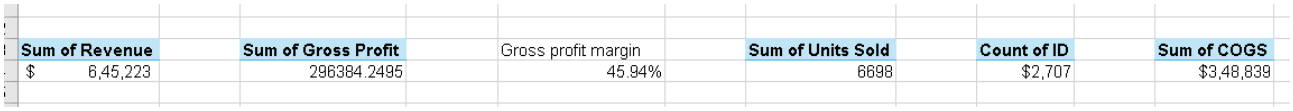


Figure 2.6 - Margin result 0.459351522 (shown as 45.94% on the dashboard)



G	H	I	J
Sum of Units Sold		Count of ID	
6698		2707	

Figure 2.7 - Units Sold (6,698) and Count of ID (2,707) KPI pivots



Sum of Revenue	Sum of Gross Profit	Gross profit margin	Sum of Units Sold	Count of ID	Sum of COGS
$ 6,45,223	296384.2495	45.94%	6698	$2,707	$3,48,839

Figure 2.8 - All KPI pivot cells together: Revenue, Gross Profit, Margin, Units, Orders, COGS

8. Timeline

Phase	Activity	Duration
Week 1	Data cleaning, helper columns, validation	1 week
Week 2	Planning, KPI definition, KPI pivots	1 week
Week 3	Charts and visual report	1 week
Week 4	Statistics and scenario models	1 week
Week 5	Dashboard, slicers, final report	1 week

9. Expected Outcomes

- A validated set of six KPIs ready for display.
- Clear view of seasonality, top markets and channel mix.
- A reusable workflow that refreshes automatically when new data is added.

Evaluation focus: originality and clarity of the plan, robustness of the KPI rationale, and thorough documentation of tools and timeline.

WEEK 3

Data Visualization and Reporting with Advanced Excel

Objective: build an informative visual report with charts and pivot charts, and explain the reasoning behind each visual.

1. Task Overview

Five visualizations were created from pivot tables on the sales dataset (simulated business data with revenue, profit, country, region and channel). Each chart answers one business question, and each chart type was chosen to match the nature of the data.

2. Visual 1: Revenue and Profit by Month (Line Chart with Markers)

Why: a line chart with markers is the standard choice for showing how two measures change over ordered time periods, making trends and seasonality easy to see.

Steps:

- Build a pivot table with Month in Rows and Sum of Revenue and Sum of Gross Profit in Values.
- Sort Sum of Revenue from Largest to Smallest to check the ranking.
- Months must be in calendar order: Row Labels dropdown -> More Sort Options -> Manual -> More Options -> untick *Sort automatically* and choose the custom list Jan, Feb ...
- Click inside the pivot -> PivotTable Analyze -> PivotChart -> Line with Markers.
- Hide all field buttons, remove gridlines and axis values, add data labels, and format as K.

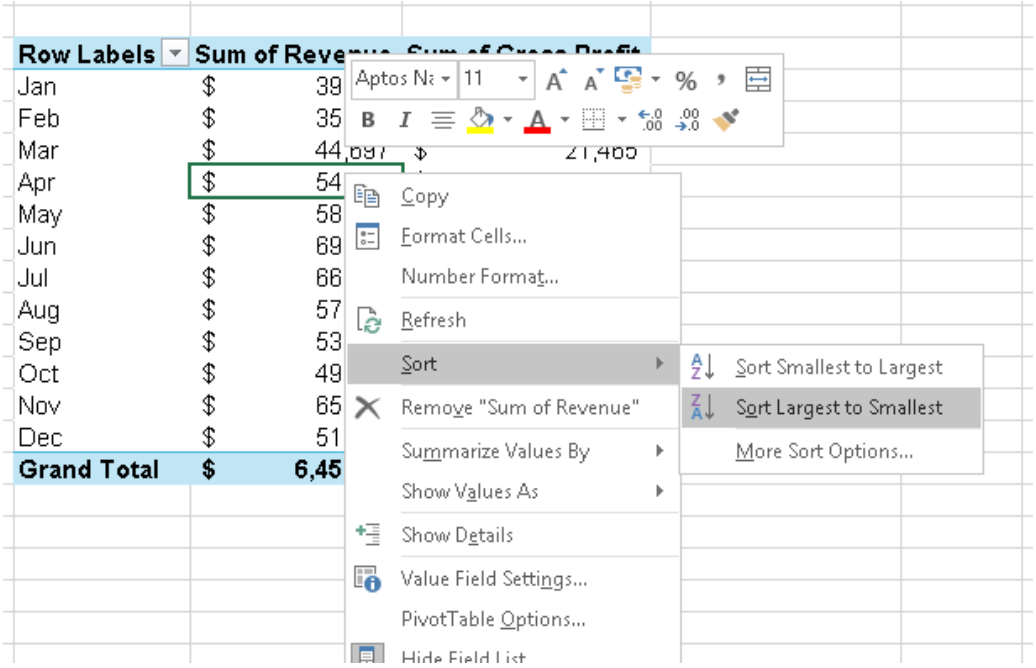


Figure 3.1 - Pivot table with monthly revenue; sorting options from the right-click menu

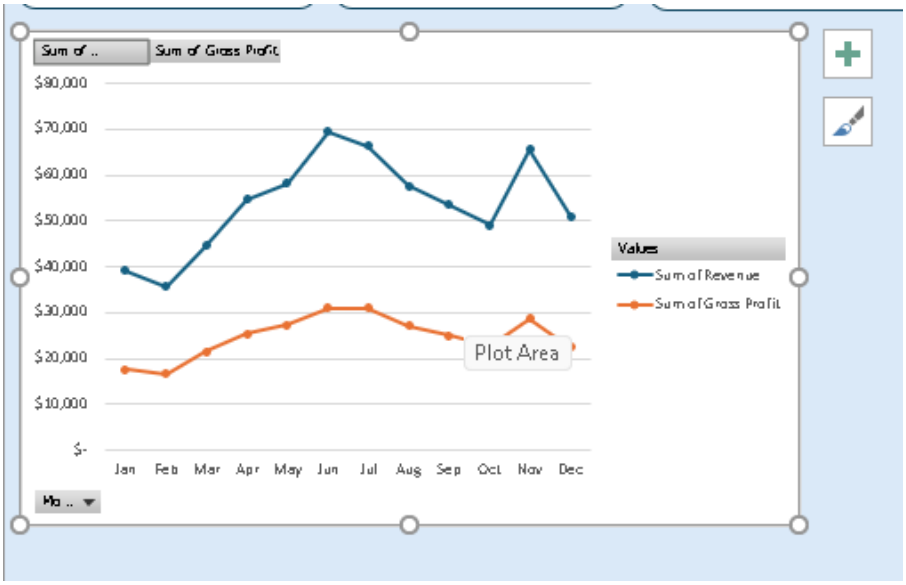


Figure 3.2 - Line chart with markers created from the pivot table

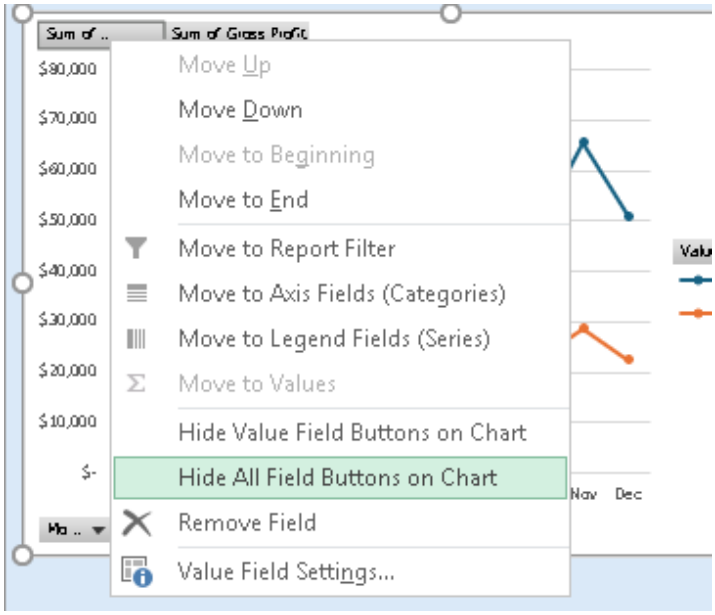


Figure 3.3 - Right-click -> Hide All Field Buttons on Chart

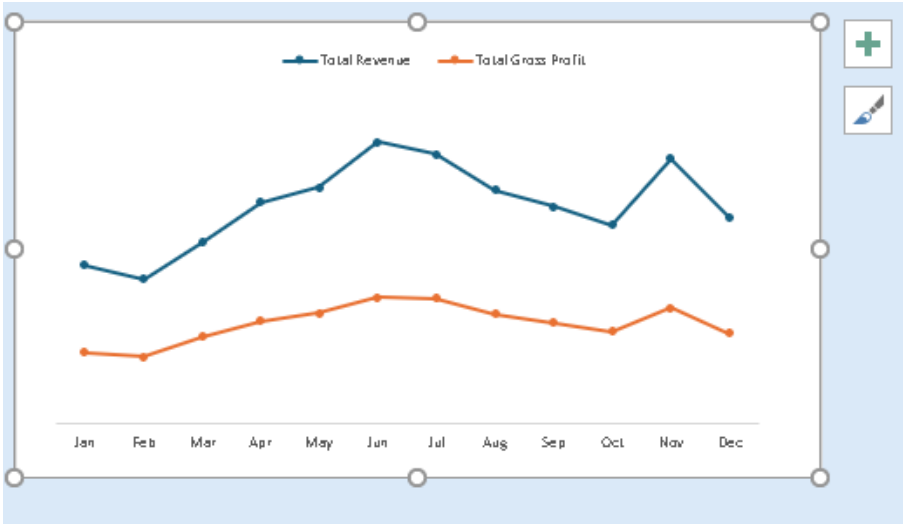


Figure 3.4 - Clean chart after removing gridlines and axis values

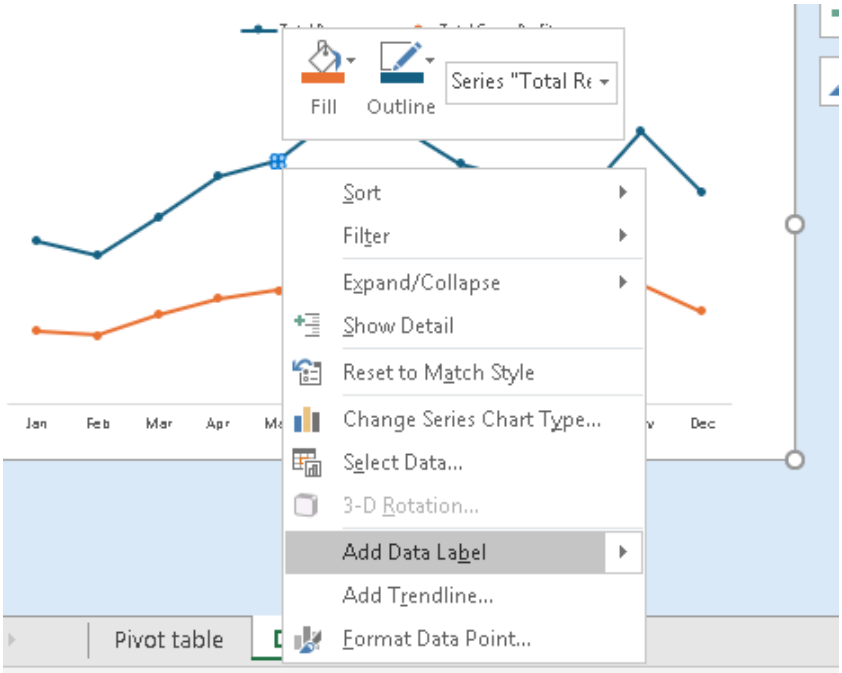


Figure 3.5 - Right-click a data point -> Add Data Label

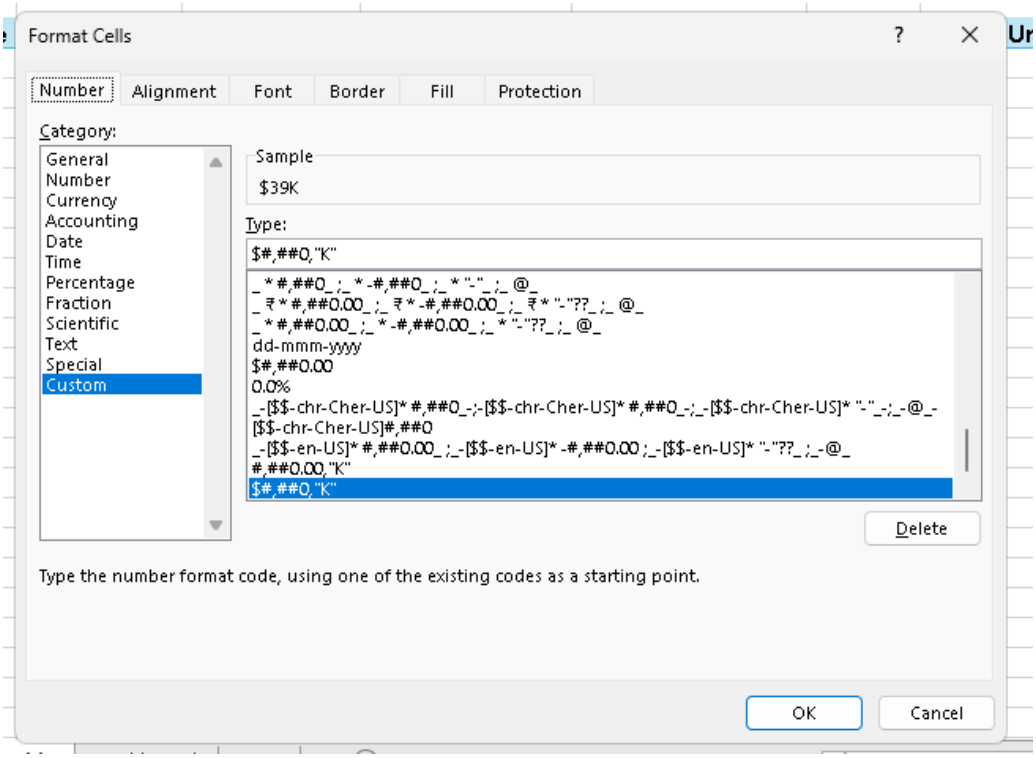


Figure 3.6 - Custom number format $#,##0,"K" shortens values to K

Row Labels	Total Revenue	Total Gross Profit
Jan	$39K	$18K
Feb	$36K	$17K
Mar	$45K	$21K
Apr	$55K	$25K
May	$58K	$27K
Jun	$70K	$31K
Jul	$66K	$31K
Aug	$58K	$27K
Sep	$54K	$25K
Oct	$49K	$23K
Nov	$66K	$29K
Dec	$51K	$22K
Grand Total	**$ 6,45,223**	**$ 2,96,384**

Figure 3.7 - Pivot values now shown as $39K, $36K ...

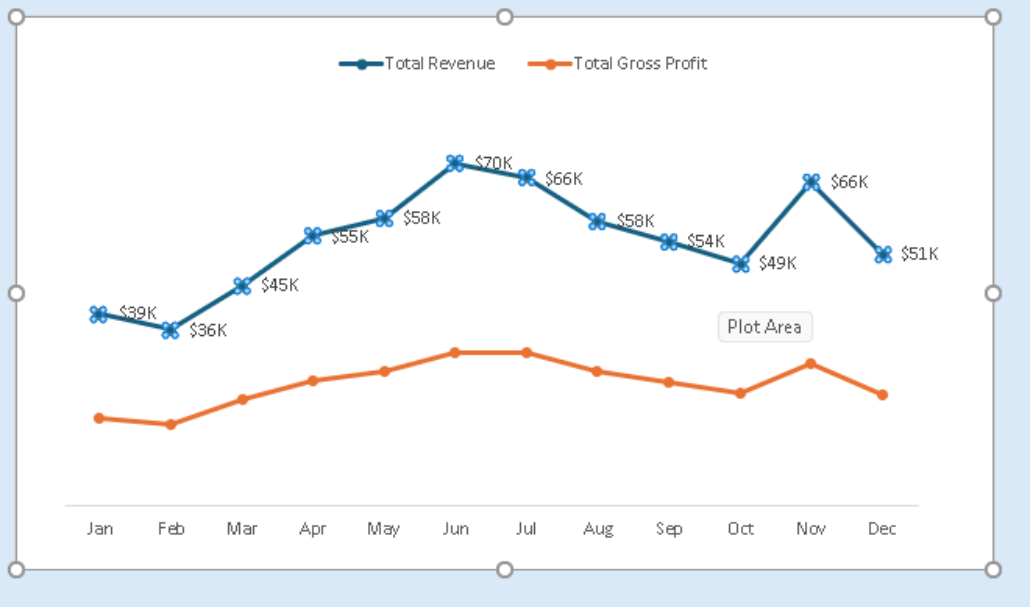


Figure 3.8 - Final line chart with Total Revenue and Total Gross Profit labels

Insight: revenue rises from about $36K in February to a peak around June (about $69K), dips through autumn and shows a second spike in October. Gross profit follows the same shape at a lower level, which means margin is stable and profit is driven mostly by volume. **Business implication:** plan stock, staff and marketing before the mid-year peak.

3. Visual 2: Top 5 Countries by Revenue (Bar Chart)

Why: a horizontal bar chart is best for ranking categories with long names, and it lets the reader compare lengths instantly.

Steps: create a pivot with Country in Rows and Total Revenue in Values; open Row Labels dropdown -> Value Filters -> Top 10 -> set **5** items by Total Revenue; insert a bar chart.

Row Labels	Total Revenue
Austria	$ 32,754
Belgium	$ 26,318
Czechia	$ 23,349
Denmark	$ 28,462
Finland	$ 24,075
France	$ 55,102
Germany	$ 70,034
Ireland	$ 16,597
Italy	$ 50,319
Netherlands	$ 42,193
Norway	$ 26,377
Poland	$ 40,290
Portugal	$ 27,591
Spain	$ 46,625
Sweden	$ 35,978
Switzerland	$ 24,732
United Kingdom	$ 74,427
Grand Total	**$ 6,45,223**

Figure 3.9 - Pivot of total revenue by all 17 countries

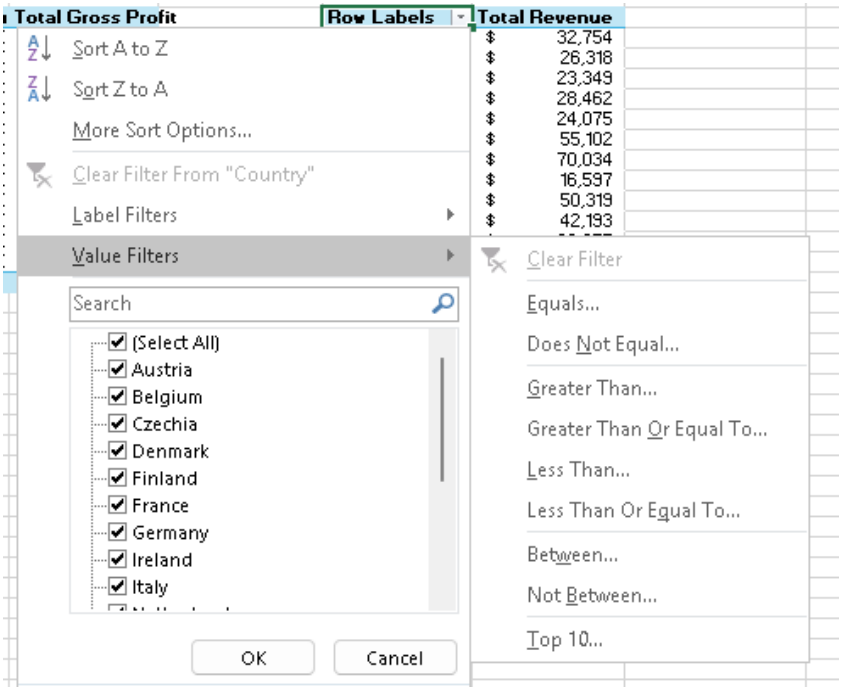


Figure 3.10 - Row Labels dropdown -> Value Filters -> Top 10

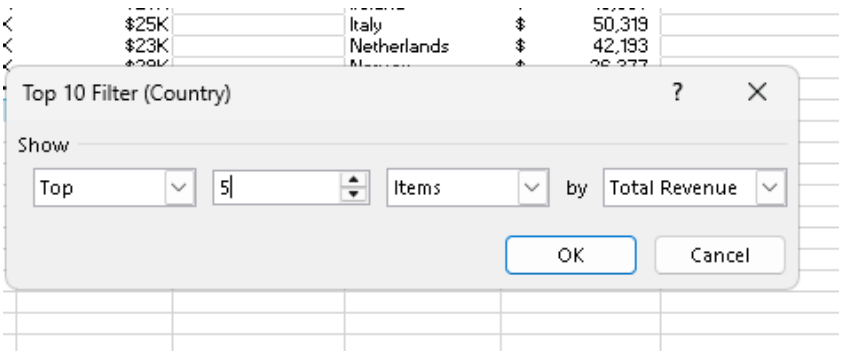


Figure 3.11 - Top 10 Filter set to Top 5 Items by Total Revenue

Row Labels	Total Revenue
France	$55K
Germany	$70K
Italy	$50K
Spain	$47K
United Kingdom	$74K
Grand Total	**$ 2,96,508**

Figure 3.12 - Filtered pivot showing the top five countries

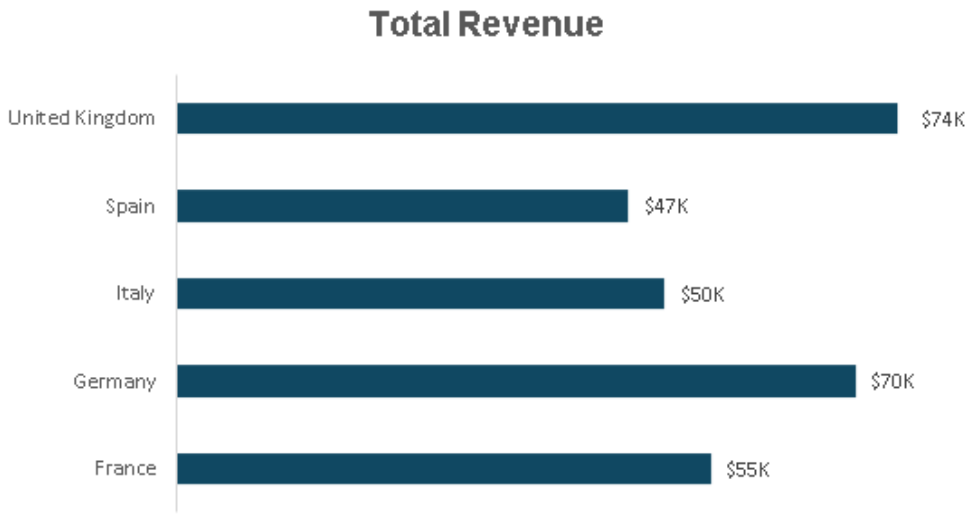


Figure 3.13 - Top 5 Countries bar chart

> **Insight:** United Kingdom ($74K), Germany ($70K), France ($55K), Italy ($50K) and Spain ($47K) together bring $296,507, about 46% of all revenue from 5 of 17 countries. **Implication:** protect these key accounts and test growth campaigns in mid-sized markets such as Poland and the Netherlands.

4. Visual 3: Revenue Share by Sales Channel (Pie Chart)

Why: a pie chart works here because there are only four channels and the question is about parts of a whole.

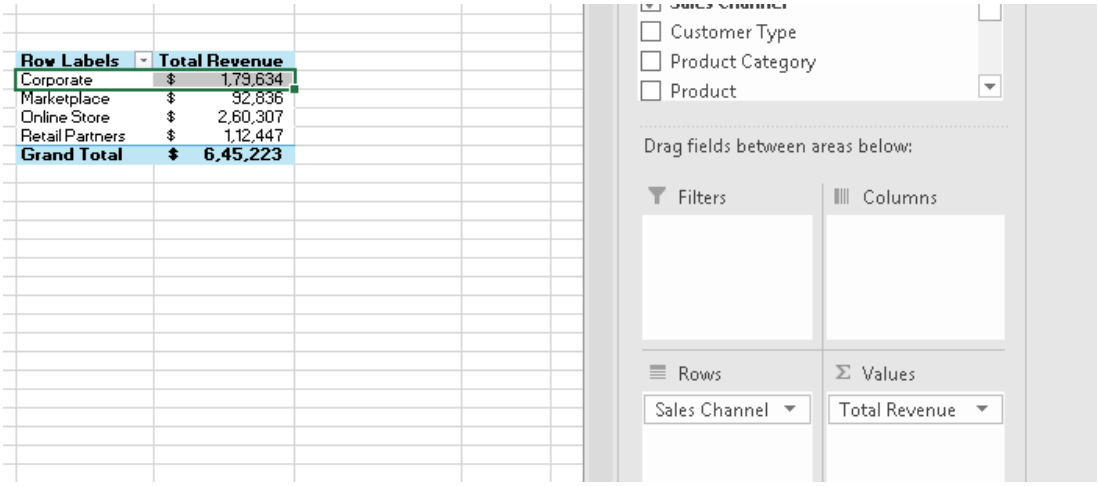


Figure 3.14 - Pivot table: Sales Channel with Total Revenue

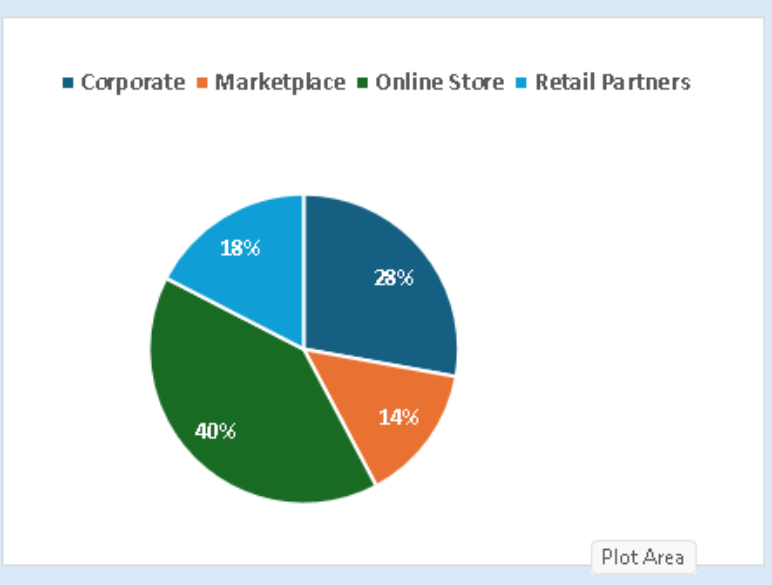


Figure 3.15 - Pie chart of revenue share by channel

Insight: the Online Store is the largest channel at about 40% (about $260K), followed by Corporate at about 28% (about $180K). Marketplace is the smallest at about 14%. **Implication:** invest in the online experience, and review whether the Marketplace channel needs promotion or should be reduced.

5. Visual 4: Region-wise Comparison (Donut Chart)

Why: the donut chart compares four regions at a glance and leaves room to add a total in the centre.

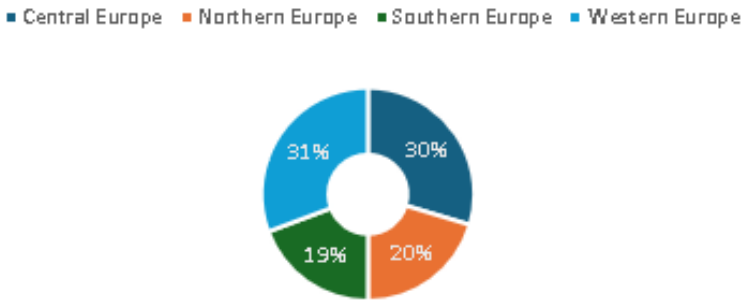


Figure 3.16 - Revenue share by region (Central, Northern, Southern, Western Europe)

Western Europe contributes the most (31%) followed by Central Europe (30%), Northern Europe (20%) and Southern Europe (19%), showing a well-balanced regional mix.

6. Visual 5: Country-wise Map Chart

Why: a filled map shows geographic spread, which a bar chart cannot. Darker shades indicate higher revenue (scale from about $16.6K to $645K in the legend).

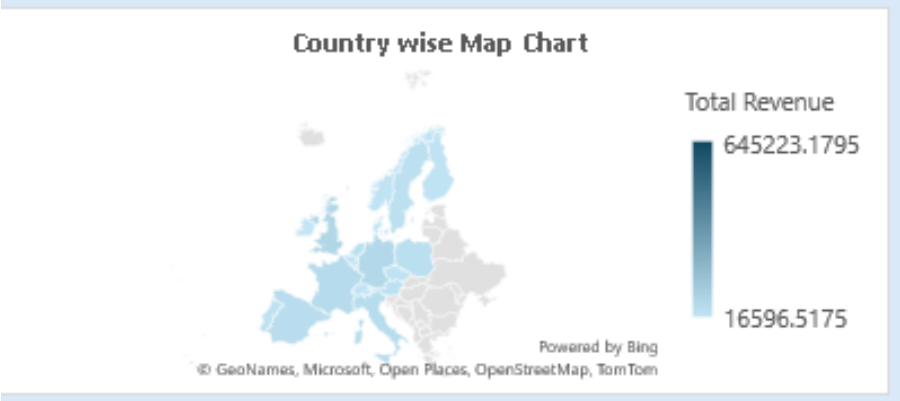


Figure 3.17 - Country-wise filled map chart (Excel Map Chart)

7. Formatting and Final Result

All charts use one colour palette (navy, orange, blue, green) so the report looks consistent; chart titles are short; field buttons, gridlines and extra axes are removed to avoid clutter.

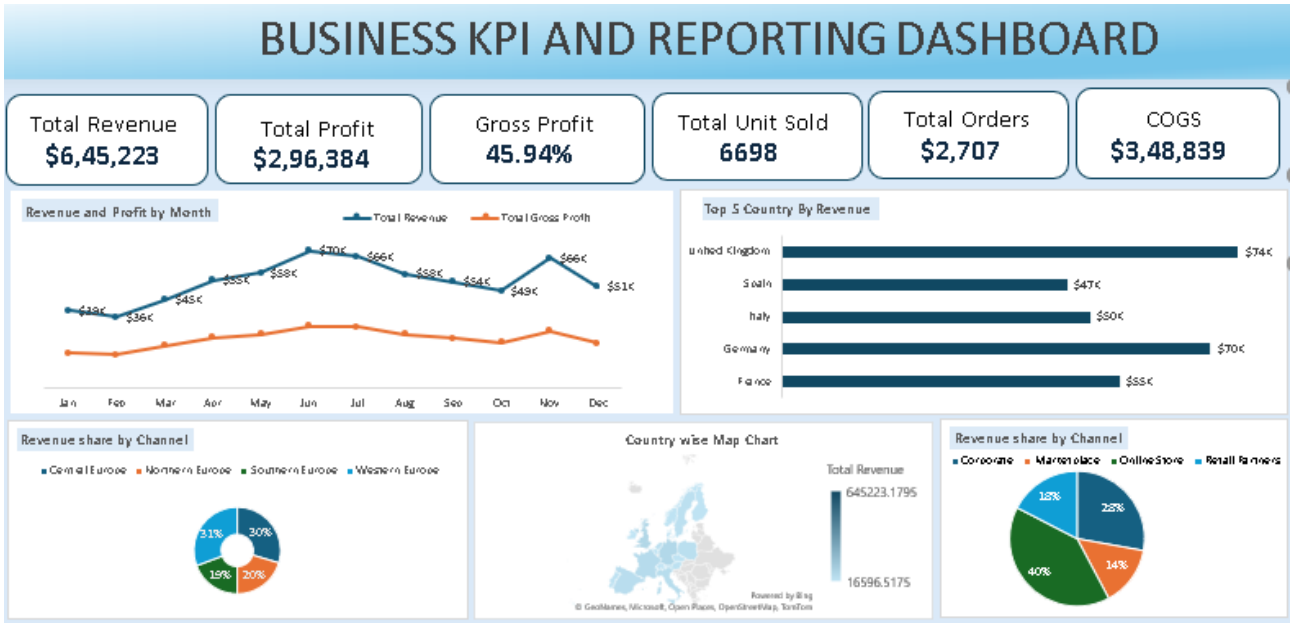


Figure 3.18 - All visuals assembled in the dashboard

8. Additional Visual Techniques (Added)

- **Sparklines** (Insert -> Sparkline -> Line) beside each month row for a tiny trend per KPI.
- **Conditional formatting** (Home -> Conditional Formatting -> Color Scales / Data Bars) in the pivot tables to highlight high and low countries.
- **Chart titles written as insights**, for example 'Online Store drives 40% of revenue', so the message is clear without explanation.

9. Overall Business Implications

- Time: seasonality peaks around mid-year.
- Place: five countries produce about 46% of revenue.
- Channel: Online Store is the main engine, so website performance is critical.
- Region: balanced, with no region dependent on one market alone.

Evaluation focus: appropriate chart choices, completeness of explanations and how directly the visuals support the stated insights.

WEEK 4

Advanced Statistical Analysis and Scenario Modeling in Excel

Objective: apply descriptive and inferential statistics and what-if scenario models to understand data dynamics.

1. Task Overview

In the role of a Junior Data Analyst, this section applies statistical methods and scenario models to the sales dataset. The statistics use the real country revenue totals from the pivot table (Figure 3.9), and the scenarios use the real KPI totals from the dashboard. Scenario percentages are assumptions chosen to represent realistic business situations.

2. Dataset Description

Item	Detail
Source	Simulated sales dataset in SalesDataTable (Excel Table)
Period	Calendar year 2026
Records	2,707 orders, 6,698 units
Dimensions	Region, Country, Sales Channel, Customer Type, Product Category, Product, Date
Measures	Units Sold, Unit Price, Discount, Unit Cost, Revenue, Gross Profit, COGS

3. Descriptive Statistics (Revenue by Country, n = 17)

Excel formulas used: AVERAGE, MEDIAN, STDEV.S, MIN, MAX, QUARTILE.INC, SKEW, KURT. The same output can be produced with Data -> Data Analysis -> Descriptive Statistics.

Statistic	Excel formula	Result
Mean	`=AVERAGE(B2:B18)`	$37,954
Median	`=MEDIAN(B2:B18)`	$32,754
Standard deviation	`=STDEV.S(B2:B18)`	$16,658
Minimum (Ireland)	`=MIN(B2:B18)`	$16,597
Maximum (United Kingdom)	`=MAX(B2:B18)`	$74,427
Q1	`=QUARTILE.INC(B2:B18,1)`	$26,318
Q3	`=QUARTILE.INC(B2:B18,3)`	$46,625
Coefficient of variation	`=STDEV.S(B2:B18)/AVERAGE(B2:B18)`	43.9%
Skewness	`=SKEW(B2:B18)`	1.01
Kurtosis	`=KURT(B2:B18)`	0.27

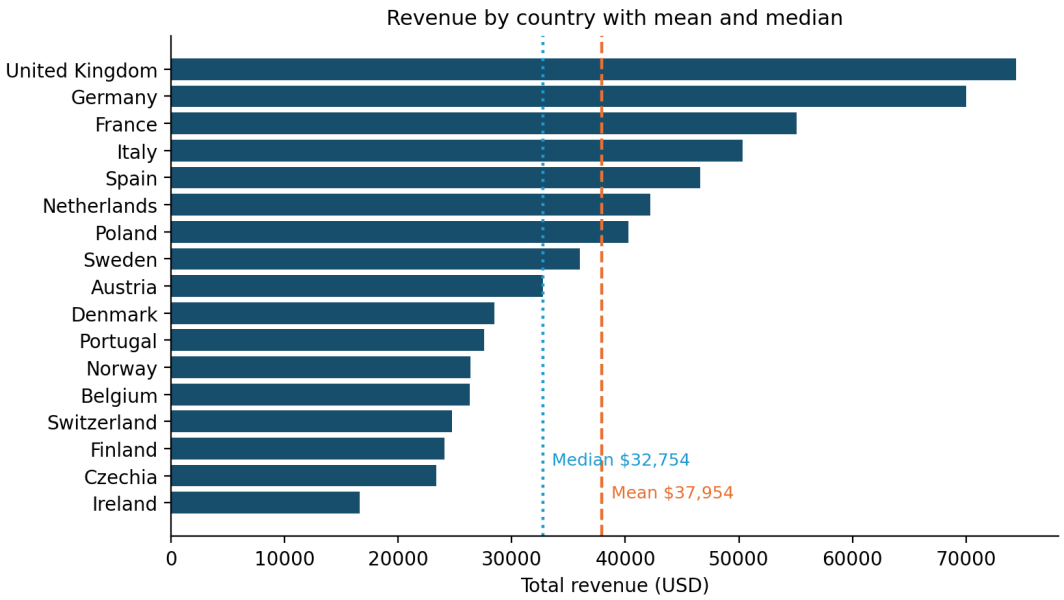


Figure 4.1 - Country revenue distribution with mean and median

Interpretation: the mean ($37,954) is higher than the median ($32,754) and the skewness is positive (1.01), so a few large markets (United Kingdom, Germany) pull the average up. The coefficient of variation of 44% shows high variation between countries, so average values should not be used to set country targets.

4. Inferential Statistics

Confidence interval for mean country revenue (95%). Using the t-distribution because n is small:

Step	Excel formula	Result
Margin of error	`=CONFIDENCE.T(0.05,STDEV.S(B2:B18),COUNT(B2:B18))`	$8,565
Lower bound	`=AVERAGE(B2:B18)-margin`	$29,390
Upper bound	`=AVERAGE(B2:B18)+margin`	$46,519

We are 95% confident that the true average country revenue lies between $29,390 and $46,519.

Regression and correlation (method). To test whether monthly gross profit moves with monthly revenue, use the monthly pivot table (Month, Revenue, Gross Profit):

Tool	Formula or path	What it tells us
Correlation	`=CORREL(Revenue_range, Profit_range)`	Strength of linear relationship (-1 to +1)
R squared	`=RSQ(Profit_range, Revenue_range)`	Share of profit variation explained by revenue
Slope / intercept	`=SLOPE(...) =INTERCEPT(...)`	Profit change per extra $1 of revenue
Full regression	Data -> Data Analysis -> Regression, or =LINEST(Y,X,TRUE,TRUE)	Coefficients, standard errors, p-values
Forecast	`=FORECAST.LINEAR(next_X, Y, X)`	Predicted profit for a planned revenue level

> Because monthly gross profit is about 46% of monthly revenue (Figure 3.8), the expected result is a very high R squared and a slope close to the margin of 0.46. Insert the screenshot of your Regression output below.

Insert screenshot: Excel Regression output (Data Analysis ToolPak) for monthly Revenue vs Gross Profit

5. Scenario Modeling (What-If Analysis)

Base case from the dashboard: revenue $645,223; COGS $348,839; gross profit $296,384; units 6,698. This gives average price = $96.33 per unit and average cost = $52.08 per unit.

Model logic (cells set up as inputs so Scenario Manager can change them):

Cell / name	Formula
Price	`=Revenue / Units (96.33)`
Units	`=Base_units * (1 + Volume_change)`
Revenue	`=Price * (1 + Price_change) * Units`
COGS	`=Unit_cost * (1 + Cost_change) * Units`
Gross Profit	`=Revenue - COGS`
Margin	`=Gross_Profit / Revenue`

Steps in Excel: Data -> What-If Analysis -> **Scenario Manager** -> Add scenario -> set the changing cells (price change, volume change, cost change) -> Show / Summary report. Use **Goal Seek** to find the volume needed to reach a target profit, and a **Data Table** for the sensitivity grid.

Scenario	Price	Volume	Unit cost	Revenue	Gross profit	Margin
Base case	+0%	+0%	+0%	$645,223	$296,384	45.9%
Optimistic	+5%	+8%	+0%	$731,683	$354,937	48.5%
Pessimistic	-3%	-10%	+4%	$563,280	$236,767	42.0%
Discount campaign	-10%	+15%	+0%	$667,806	$266,641	39.9%
Cost inflation	+0%	+0%	+6%	$645,223	$275,454	42.7%

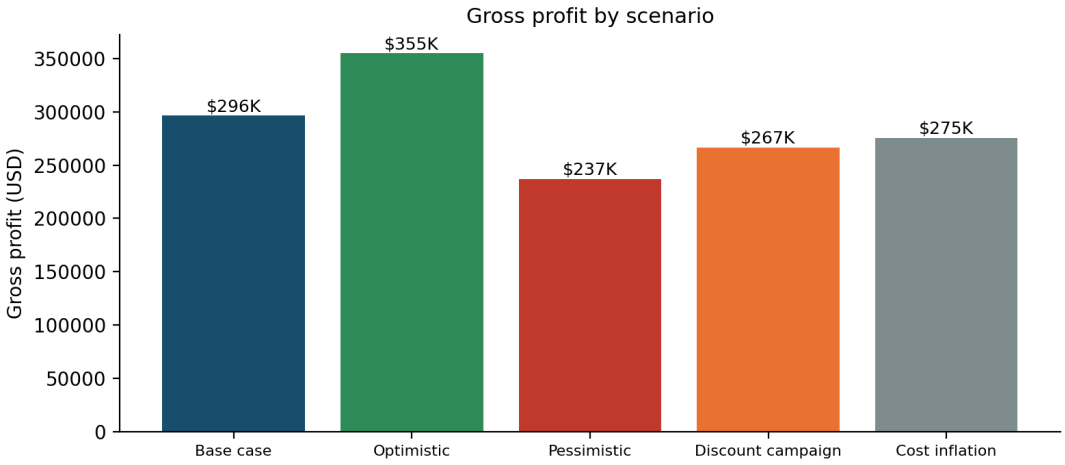


Figure 4.2 - Gross profit under each scenario

- **Optimistic:** a 5% price rise with 8% more units lifts gross profit by $58,553 (19.8%).

- **Pessimistic:** a 3% price cut, 10% fewer units and 4% higher cost reduce gross profit by $59,618 (20.1%).
- **Discount campaign:** 10% lower price with 15% more units changes gross profit by -10.0% and margin falls to 39.9%, so extra volume does not pay for the discount.
- **Cost inflation:** a 6% unit cost increase alone reduces gross profit by $20,930 (7.1%).

6. Sensitivity Table (Two-Variable Data Table)

Gross profit for combinations of price change (columns) and volume change (rows), unit cost unchanged. Green cells are above the base gross profit; red cells are below it.

Volume \ Price	-10%	-5%	+0%	+5%	+10%
-10%	$208,676	$237,711	$266,746	$295,781	$324,816
-5%	$220,269	$250,917	$281,565	$312,213	$342,861
+0%	$231,862	$264,123	$296,384	$328,645	$360,906
+5%	$243,455	$277,329	$311,203	$345,078	$378,952
+10%	$255,048	$290,535	$326,023	$361,510	$396,997

Reading the table: price changes matter more than volume changes. A 10% price cut needs more than 10% extra volume just to break even on gross profit, because every unit is already earning about $44 on a $96 price. Break-even volume for a 10% price cut = 27.8% more units.

7. Linking Statistics to Action

- Revenue is skewed to a few countries, so set targets per country using each country's own mean and variation, not a global average.
- Margin protection (price and cost control) has a larger effect on profit than discount-driven volume.
- Use the Optimistic, Base and Pessimistic range ($ gross profit) as the planning band for budgeting.

Evaluation focus: correct statistical functions, depth (descriptive, inferential, regression method, scenarios, sensitivity) and relevance of insights to business decisions.

WEEK 5

Dashboard Development and Final Reporting in Excel

Objective: consolidate the analysis into a fully interactive dashboard and document how it was built and how it supports decisions.

1. Task Overview

The final dashboard, titled **Business KPI and Reporting Dashboard**, brings together all KPIs and charts on one sheet. It is built from pivot tables and pivot charts connected to slicers, so every number and chart responds to the same filter.

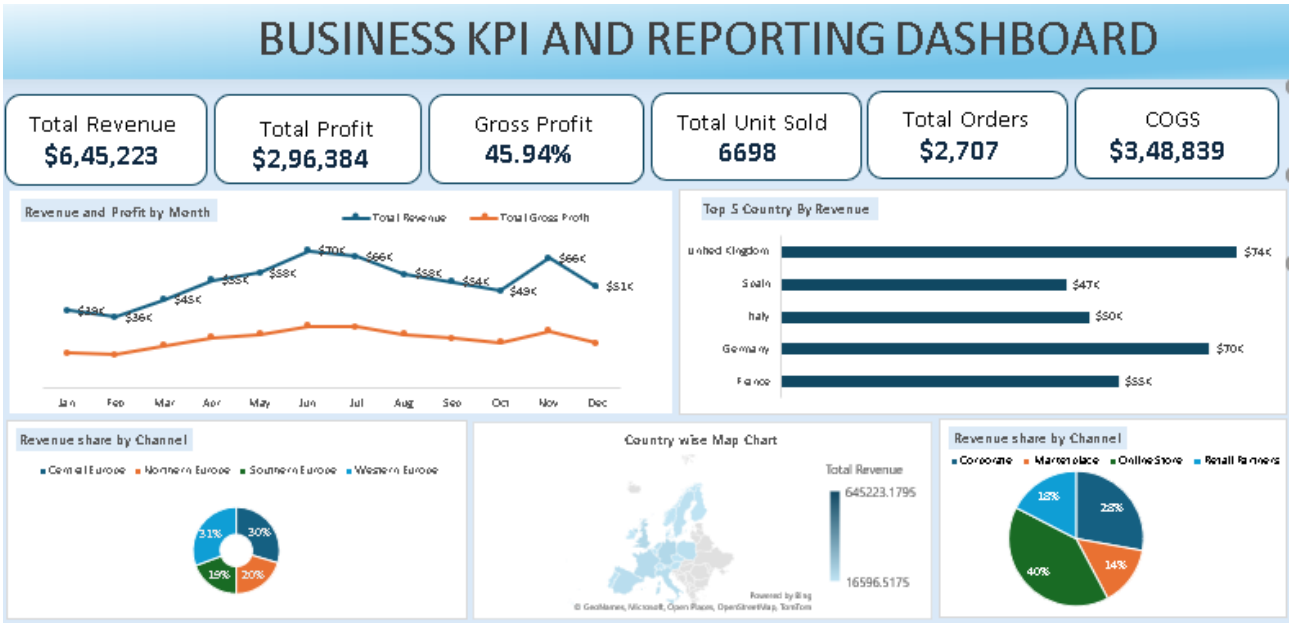


Figure 5.1 - Final Business KPI and Reporting Dashboard

2. Data Assumptions

- Data is the simulated SalesDataTable (2,707 orders, 17 countries, 4 regions, 4 channels, calendar year 2026).
- Revenue is shown in USD; gross profit = revenue - COGS; margin = gross profit / revenue.
- Helper columns (Year, Month No, Month, Quarter) from Week 1 are used for time analysis.
- All pivot tables use the same table, so a single Refresh All updates the dashboard.

3. Step-by-Step Build Guide

Step	Action	Detail
1	Prepare the data	Convert to Excel Table, add Year, Month No, Month, Quarter columns (Week 1)
2	Create KPI pivot tables	Insert -> PivotTable for Revenue, Gross Profit, Units Sold, Count of ID, COGS
3	Calculate margin	`=C4/A4` formatted as a percentage
4	Create a new Dashboard sheet	Hide gridlines (View -> untick Gridlines); add a title banner

Step	Action	Detail
5	Design KPI cards	Insert rounded rectangles; link each to a KPI cell by typing = in the formula bar and clicking the cell
6	Build charts	Line, Top 5 bar, channel pie, region donut, map chart (Week 3)
7	Apply formatting	Consistent colours, short titles, K-format data labels
8	Add slicers	PivotTable Analyze -> Insert Slicer -> Product Category
9	Connect slicers	Slicer -> Report Connections -> select all pivot tables
10	Test and align	Click each slicer item and confirm every chart changes

4. Key Dashboard Components

KPI cards. Six cards sit across the top: Total Revenue $6,45,223; Total Profit $2,96,384; Gross Profit Margin 45.94%; Total Unit Sold 6,698; Total Orders 2,707; COGS $3,48,839. Each card is a shape linked to a pivot cell, so it updates when the slicer changes.

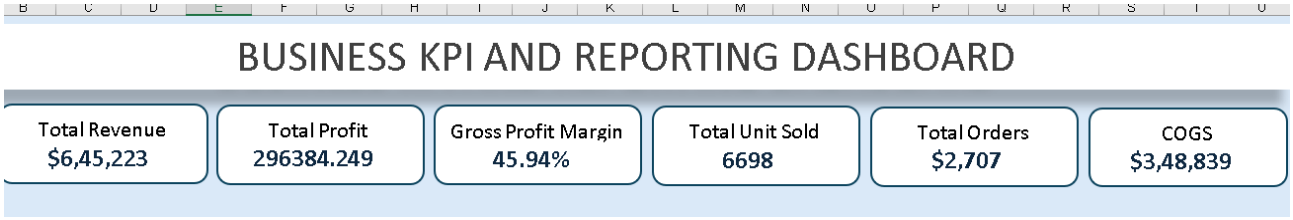


Figure 5.2 - KPI cards on the dashboard

Charts. The dashboard layout follows a logical top-to-bottom order: totals first (KPI cards), then time trend (revenue and profit by month), then ranking (top 5 countries), then composition (region donut, map, channel pie).

5. Interactive Elements

A **Product Category slicer** (Camping, Cycling, Fitness, Hiking, Travel) was added. At first, clicking a category did not change the charts because the slicer was linked only to its own pivot table. This was fixed with **Slicer -> Report Connections**, ticking every pivot table so all charts respond together.

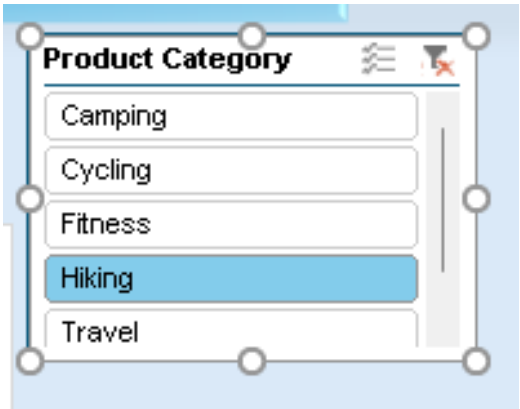


Figure 5.3 - Product Category slicer (Hiking selected)

Element	How it contributes to analysis
Product Category slicer	Compare categories instantly; every KPI, chart and map updates
Report Connections	Keeps all visuals in sync so numbers never conflict

Element	How it contributes to analysis
KPI cards	Immediate headline view for executives
Pivot charts	Drill down and re-slice without rebuilding the chart
Map chart	Spots geographic strengths and gaps

6. Design Rationale

- **Top-to-bottom reading order:** summary, then trend, then detail.
- **Single colour family** (blue shades with orange accent) keeps focus on data.
- **K-format and no gridlines** remove clutter.
- **Different chart types** match different questions: line for time, bar for ranking, pie and donut for share, map for location.

7. Potential Improvements (Added)

- Add a **Timeline slicer** for Order Date and slicers for Region, Sales Channel and Customer Type.
- Add a **dynamic title** that shows the selected category, for example ="Revenue for "&slicer;_cell.
- Use a **Data Validation dropdown** to switch the KPI shown on the chart (Revenue, Profit or Units) with CHOOSE or INDEX/MATCH.
- Add **year-over-year and target vs actual** indicators once more years of data are available.
- Protect the dashboard sheet and hide the pivot sheets to prevent accidental changes.
- Publish to Power BI or SharePoint for broader sharing.

8. Reflective Summary: Decision-Making

The dashboard lets managers answer key questions in seconds:

- **Where do we earn most?** UK, Germany, France, Italy and Spain deliver about 46% of revenue, so account management should prioritise them.
- **Which channel should get investment?** The Online Store contributes about 40% of revenue.
- **Is profit healthy?** A 45.94% gross margin shows strong unit economics; any scenario that cuts price needs a matching volume gain (Week 4).
- **When should we prepare?** Revenue peaks mid-year, so stock and campaigns should be planned in advance.
- **Which product categories perform best?** The Product Category slicer reveals this by filtering the whole dashboard.

9. Conclusion

Across five weeks, raw sales data was cleaned and structured (Week 1), a KPI-driven analysis plan was created (Week 2), insights were visualized (Week 3), validated with statistics and scenarios (Week 4) and finally delivered as an interactive dashboard (Week 5). The project shows a complete analyst workflow from data to decision.

> **Evaluation focus:** dashboard functionality and creativity, clarity of the build guide and a clear connection from dashboard features to business insights.